

⚡ The Laplace Transform (Probability Generating Function)

The function you refer to, $g(t) = E[e^{-tX}]$, is formally known as the **Laplace Transform** of the density function $f(x)$. In the context of probability, for non-negative, integer-valued random variables (like the discrete distributions), it is often referred to as the **Probability Generating Function (PGF)** when using z instead of e^{-t} .

1. Definition and Relationship to MGF

The Laplace Transform $g(t)$ is simply the MGF $\phi(t)$ evaluated at a negative argument:

$$g(t) = E[e^{-tX}] = \phi_X(-t)$$

Function	Argument Transformation
MGF $\phi(t)$	e^{tX}
Laplace Transform $g(t)$	e^{-tX}

2. The Key Advantage (Guaranteed Existence)

The primary mathematical advantage of the Laplace Transform (or PGF) is its guaranteed existence for all non-negative random variables $X \geq 0$.

Function	Existence Issue	Behavior for $X \geq 0, t > 0$
MGF $\phi(t) = E[e^{tX}]$	The term e^{tX} grows exponentially fast. The integral/sum may diverge (go to ∞).	$e^{tX} \geq 1$ (potentially very large)
Laplace Transform $g(t) = E[e^{-tX}]$	The term e^{-tX} decays exponentially fast. The integral/sum is always finite .	$0 \leq e^{-tX} \leq 1$

This stability makes the Laplace Transform the preferred tool in fields like **queuing theory**, **reliability analysis**, and **stochastic processes**, which deal almost exclusively with non-negative waiting times, counting processes, and lifetimes.

3. Alternative Use: The PGF (Probability Generating Function)

For **discrete, non-negative integer-valued** random variables ($X \in \{0, 1, 2, \dots\}$), the Laplace Transform is often written as the PGF:

$$G_X(z) = E[z^X]$$

- **Relationship:** By letting $z = e^t$ (or $z = e^{-t}$ depending on convention), the PGF is directly related to the MGF.
- **Moment Trick:** The PGF also generates moments, but with a slight difference: $E[X] = G'_X(1)$, $E[X(X - 1)] = G''_X(1)$, etc. It's often easier for discrete problems.

2. ⚡ Binomial MGF Derivation: A Masterclass in Factoring

The derivation of the Binomial MGF is crucial because it demonstrates the power of pattern recognition in MGF calculations—specifically, recognizing the **Binomial Theorem**.

Goal: Find $\phi_X(t)$ for $X \sim \text{Binomial}(n, p)$

The PMF is $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.

Step 1: Start with the MGF Definition (LOTUS)

Since X is discrete, we use the summation form:

$$\phi(t) = E[e^{tX}] = \sum_{k=0}^n e^{tk} \cdot P(X = k)$$

Step 2: Substitute the PMF

Substitute the Binomial probability formula into the MGF expression:

$$\phi(t) = \sum_{k=0}^n e^{tk} \cdot \binom{n}{k} p^k (1-p)^{n-k}$$

Step 3: Group Terms by Exponent *k*

The key step is to combine e^{tk} and p^k . Since $e^{tk}p^k = (e^t)^k p^k = (pe^t)^k$:

$$\phi(t) = \sum_{k=0}^n \binom{n}{k} (pe^t)^k (1-p)^{n-k}$$

Step 4: Apply the Binomial Theorem

The **Binomial Theorem** states:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

We observe a perfect match with:

- $a = pe^t$
- $b = 1 - p$

Therefore, the entire sum collapses into the binomial expansion of $(a + b)^n$:

$$\phi(t) = (\underbrace{pe^t}_a + \underbrace{1-p}_b)^n$$

Final Result

$$\phi(t) = (1 - p + pe^t)^n$$

This result is important because it validates the MGF **Superpower 1** (MGF of sum is product of MGFs): A Binomial variable is the sum of n independent Bernoulli variables, and this MGF is exactly the Bernoulli MGF $\phi_{Bern}(t) = (1 - p + pe^t)$ raised to the power of n .

3. Summary of Key MGF Properties

Your final notes accurately summarize why MGFs are so valuable:

A. The Two Superpowers

1. **Sums of Independent RVs:** For $Z = X + Y$, the distribution of Z can be found simply by multiplying the MGFs:

$$\phi_{X+Y}(t) = \phi_X(t) \cdot \phi_Y(t)$$

2. **Uniqueness Property:** The MGF is the distribution's unique algebraic signature (fingerprint). If you know the MGF, you know the distribution.

B. Quick Reference Table

Distribution	MGF $\phi(t)$
Poisson (λ)	$e^{\lambda(e^t-1)}$
Exponential (λ)	$\frac{\lambda}{\lambda-t}$

Distribution	MGF $\phi(t)$
Normal (μ, σ^2)	$e^{\mu t + \sigma^2 t^2 / 2}$

Would you like to explore an example of using the Laplace Transform for a non-negative continuous random variable, such as finding the mean of the Exponential distribution using $g(t) = E[e^{-tX}]$ instead of $\phi(t) = E[e^{tX}]$?